



Diagnosis of primary aldosteronism

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INTRODUCTION

Autonomous (primary) hypersecretion of [aldosterone](#) is an underdiagnosed cause of [hypertension](#). The classic presenting signs of primary aldosteronism are [hypertension](#) and [hypokalemia](#), but [potassium](#) levels are frequently normal in modern-day series of primary aldosteronism. The most common subtypes of primary aldosteronism are:

- A unilateral [aldosterone-producing adenoma](#) (APA; ≥ 10 mm) or [aldosterone-producing micronodule](#) (< 10 mm)
- Bilateral idiopathic hyperaldosteronism (IHA; bilateral [adrenal hyperplasia](#))

Less common forms include:

- Familial hyperaldosteronism (FH) types I to IV and primary aldosteronism with [seizures](#) and neurologic abnormalities (PASNA) (see "[Familial hyperaldosteronism](#)")
- Unilateral [hyperplasia](#) or primary [adrenal hyperplasia](#) (caused by micronodular or macronodular [hyperplasia](#) of the [zona glomerulosa](#) of one [adrenal gland](#))
- Pure [aldosterone-producing adrenocortical carcinomas](#)
- Ectopic [aldosterone-producing tumors](#)

The diagnosis of primary aldosteronism will be reviewed here. The clinical manifestations and treatment of this disorder are discussed separately. (See "[Pathophysiology and clinical features of primary aldosteronism](#)" and "[Treatment of primary aldosteronism](#)".)



PREVALENCE AND CLINICAL FEATURES



Prevalence — Older studies suggested a [prevalence](#) of primary aldosteronism of less than 1 percent of patients with [hypertension](#). However, subsequent studies document a considerably higher [prevalence](#) [1-4]. In a retrospective, multicenter review, more widespread measurement of the plasma [aldosterone](#) concentration (PAC) and plasma [renin](#) activity (PRA) as a case-detection test in patients with [hypertension](#) resulted in marked increases (1.3- to 6.3-fold) in the annual detection rate of primary aldosteronism and in the [proportion](#) of patients

with [hypertension](#) in whom primary aldosteronism was detected (1 to 2 percent before screening versus 5 to 10 percent after screening) [1,3].

In addition, in a [cross-sectional study](#), a continuum of [renin-independent aldosterone](#) production was evident within every blood pressure category; greater [renin-independent aldosterone](#) production was associated with higher blood pressure, greater kaliuresis, and lower serum [potassium](#) levels [5]. The adjusted [prevalence](#) estimates for primary aldosteronism were 11.3 percent in patients with normotension and 15.7, 21.6, and 22 percent in those with stage 1, stage 2, and [resistant hypertension](#), respectively. Thus, primary aldosteronism appears to be highly prevalent and underrecognized [5]. Consequently, the 2025 Endocrine Society clinical practice guideline suggests that all individuals with [hypertension](#) be screened for primary aldosteronism [6]. (See '[Patient selection](#)' below.)

Variable presentation — The presence of primary [mineralocorticoid](#) excess should be highly suspected in any patient with the triad of [hypertension](#), unexplained [hypokalemia](#), and [metabolic alkalosis](#) [7-9]. However, most patients with primary [mineralocorticoid](#) excess have normokalemia and, rarely, some have [hypokalemia](#) but normotension (primarily young female adults) [10]. In patients who present with [hypokalemia](#) and normal blood pressure, surreptitious vomiting, [diuretic](#) therapy, and [Bartter syndrome](#) should be excluded; these disorders lead to [secondary hyperaldosteronism](#) with increased PRA and direct [renin](#) concentration (DRC) rather than the suppressed PRA and DRC values evident in primary aldosteronism. (See '[Other causes of hypertension and hypokalemia](#)' below.)

An estimated 9 to 37 percent of patients with primary aldosteronism have [hypokalemia](#) [1]. This reduction in [prevalence](#) likely reflects earlier diagnosis, as more patients with [hypertension](#) are being screened for primary aldosteronism. (See "[Pathophysiology and clinical features of primary aldosteronism](#)", section on '[Hypokalemia: An inconsistent finding](#)'.)

Patients with the rare genetic disorder [glucocorticoid-remediable aldosteronism](#) (GRA) generally present with normokalemia. The pathophysiology of GRA and other forms of familial hyperaldosteronism is reviewed elsewhere. (See "[Familial hyperaldosteronism](#)".)

Diagnostic delay — Low rates of testing lead to both missed diagnoses and diagnostic delay. In an international survey of 684 individuals with primary aldosteronism, over one-third of respondents reported that at least five years had elapsed between their diagnoses of [hypertension](#) and primary aldosteronism [11]. These findings underscore the need for additional provider education about the importance of testing, the increased kidney and cardiovascular [morbidity](#) from [mineralocorticoid](#) excess, and the possibility of preventing this [morbidity](#) with prompt diagnosis and appropriate treatment. (See "[Treatment of primary aldosteronism](#)", section on '[Treatment goals](#)'.)



CASE DETECTION

Case-detection testing with measurement of plasma [aldosterone](#) concentration (PAC) and [renin](#) (plasma [renin](#) activity [PRA] or direct [renin](#) concentration [DRC]) is suggested for all individuals with [hypertension](#) [6]. As noted, the [prevalence](#) of primary aldosteronism in patients with [hypertension](#) is considerably higher than previously thought. In addition, more widespread testing has demonstrated that normokalemic, rather than [hypokalemic](#),

hypertension is the most common presentation of primary aldosteronism [1,12,13]. (See ['Prevalence'](#) above and ['Variable presentation'](#) above.)

Our approach, outlined below, is consistent with the 2025 Endocrine Society clinical practice guidelines [6]. Recommendations for the treatment of primary aldosteronism are reviewed separately. (See ["Treatment of primary aldosteronism"](#).)

Patient selection — The author of this topic agrees with the 2025 Endocrine Society clinical practice guidelines, which advise that all individuals with hypertension should be tested for primary aldosteronism based on the high prevalence of primary aldosteronism, its underdiagnosis, and the potential to reduce cardiovascular morbidity and mortality through targeted treatment [6]. However, some experts endorse a targeted approach to evaluating patients for secondary causes of hypertension, including primary aldosteronism. (See ["Evaluation of secondary hypertension"](#) and ["Overview of hypertension in adults"](#) and ["Hypertension in adults: Blood pressure measurement and diagnosis"](#), section on ['Additional evaluation and follow-up'](#).)

Identifying primary aldosteronism is important because of both its high prevalence and association with higher cardiovascular morbidity and mortality compared with primary hypertension at the same degree of blood pressure elevation [14]. In patients diagnosed with primary aldosteronism, treatment of the mineralocorticoid excess results in reversal or improvement of hypertension and remediation of the increased cardiovascular risk. Nonetheless, clinical data indicate that testing for primary hyperaldosteronism is performed in only approximately 2 to 4 percent of patients with known high prevalence for this disorder, including those with resistant hypertension and hypokalemia [15-18]. Patients who are tested have higher rates of mineralocorticoid receptor antagonist (MRA) use and better blood pressure control over time [15].

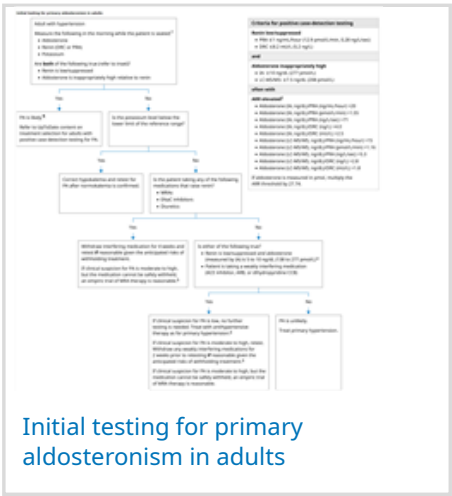
Initial testing — For initial testing, paired measurements of the PRA (or DRC), plasma aldosterone concentration (PAC), and serum potassium should be obtained in the morning in a seated ambulatory patient [6-9]. Some clinicians calculate an aldosterone-to-renin (eg, PAC/PRA) ratio as part of the case-detection strategy, but we prefer to use the paired random PAC and PRA (or DRC). Renin can be measured in terms of its enzymatic activity (PRA) or its mass (DRC) [19]. Details about PRA and DRC measurements are reviewed separately. (See ["Assays of the renin-angiotensin-aldosterone system"](#), section on ['Renin'](#).)

Blood samples should preferably be collected in the morning (eg, 8 AM) from a seated, ambulatory patient ([algorithm 1](#)). Most antihypertensive medications can be continued during biochemical testing, and posture stimulation is not required [20-25]. Although some antihypertensive drugs may affect these laboratory measurements, most drugs may be continued during initial case-

selection testing. If initial testing is negative and medication interference is suspected, management is reviewed below. (See [negative findings on initial testing'](#) below.)

Test interpretation — Patients with primary aldosteronism typically have the following findings:

- **Very low renin** – The PRA and DRC are typically very low, due in part to the associated mild volume expansion. The PRA is usually



Algorithm 1 - larger image below

≤ 1 ng/mL/hour (0.2778 ng/L per sec), and the DRC is below the lower limit of normal (typically ≤ 8.2 mU/L) ([algorithm 1](#)) [26].

Conversely, an increased PRA or DRC in a patient with hypokalemia and hypertension is most often due to diuretic therapy, renovascular or malignant hypertension, or, rarely, a renin-secreting tumor. (See '[Other causes of hypertension and hypokalemia](#)' below.)

- **Elevated aldosterone** – The PAC is usually ≥ 10 ng/dL (277 pmol/L) when measured by immunoassay or ≥ 7.5 ng/dL (208 pmol/L) when measured by liquid chromatography-tandem mass spectrometry (LC-MS/MS).

For patients with case-detection testing results that are consistent with primary aldosteronism, subsequent evaluation is reviewed below. (See '[Positive findings on initial testing](#)' below.)

If testing results are **not** consistent with primary aldosteronism, potential causes of false-negative results should be considered. (See '[Negative findings on initial testing](#)' below.)

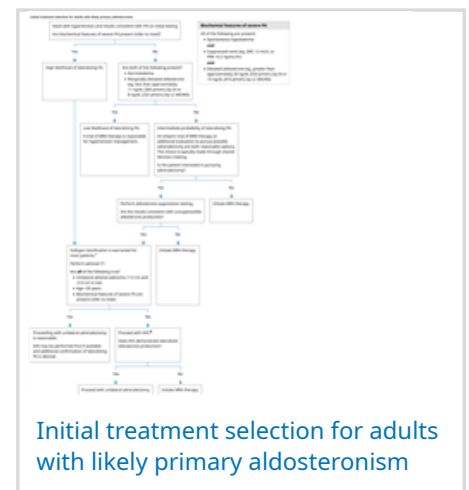
Aldosterone-to-renin ratio — Although some clinicians use an aldosterone-to-renin (eg, PAC/PRA) ratio as part of the case-detection strategy [27], this ratio is influenced by several laboratory- and assay-related variables. Therefore, we prefer to use the absolute PAC and PRA values for case-detection testing ([algorithm 1](#)).

A PAC/PRA ratio >20 ng/dL per ng/mL/hour (>555 pmol/L per ng/mL/hour) when PAC is measured by immunoassay or >15 ng/dL per ng/mL/hour (>416 pmol/L per ng/mL/hour) when PAC is measured by LC-MS/MS suggests primary aldosteronism ([algorithm 1](#)) [28]. Cutoff values depend on whether hormone concentrations are expressed in conventional or SI units, and they vary with the PRA assay. The lower limit of detection varies among the different PRA assays and can have a dramatic effect on the PAC/PRA ratio [29]. For example, a very different ratio is obtained if the lower limit of detection for PRA is 0.6 ng/mL/hour compared with 0.1 ng/mL/hour; for a PAC of 16 ng/dL (444 pmol/L), the PAC/PRA ratio would be 27 and 160, respectively. A significant number of patients with hypertension have a low renin concentration, but PAC is normal. A high PAC (typically ≥ 10 ng/dL [277 pmol/L] by immunoassay or ≥ 7.5 ng/dL [208 pmol/L] when measured by LC-MS/MS) that is inappropriate for the corresponding renin value is uncommon in the absence of primary aldosteronism.

Subsequent evaluation

Positive findings on initial testing — In patients with findings consistent with primary aldosteronism on initial testing, the likelihood of lateralized aldosterone production (ie, aldosterone-producing adenoma [APA]) should be assessed. In addition, the potential impact of interfering medications should be considered ([algorithm 2](#)).

In patients with suspected familial hyperaldosteronism, genetic testing should also be performed. Indications for genetic testing include age <20 years and a family history of early-onset hypertension or primary aldosteronism. (See "[Familial hyperaldosteronism](#)", section on '[Approach to evaluating suspected familial hyperaldosteronism](#)'.)



• Assessing the likelihood of APA (lateralizing aldosteronism)

- **High probability** – Patients with hypokalemia, very low renin (eg, PRA <0.2 ng/mL/hr), and high PAC (eg, >20 ng/dL [>554 pmol/L] by immunoassay or >15 ng/dL [>416 pmol/L] by LC-MS/MS) have an unequivocal diagnosis of primary aldosteronism. In this clinical setting, no other diagnosis can explain the findings. If such patients would consider surgical management, they do **not** require additional diagnostic testing before proceeding to subtype classification ([algorithm 2](#)). Subtype classification is needed to distinguish APA, or rarely, carcinoma or unilateral hyperplasia, from bilateral adrenal hyperplasia (idiopathic hyperaldosteronism [IHA]). (See '[Subtype classification](#)' below.)

In patients who are not surgical candidates or would not consider surgery, MRA therapy should be initiated. (See "[Treatment of primary aldosteronism](#)", section on '[Medical therapy](#)'.)

- **Low probability** – Patients with both of the following features have a low likelihood of lateralized aldosterone production:
 - Normokalemia
 - Marginally elevated aldosterone (eg, less than approximately 11 ng/dL [305 pmol/L] by immunoassay or 8 ng/dL [222 pmol/L] by LC-MS/MS)

These patients do not generally require further evaluation. In such patients, an empiric trial of MRA therapy is reasonable. (See "[Treatment of primary aldosteronism](#)", section on '[First line: Mineralocorticoid receptor antagonists](#)'.)

- **Intermediate probability** – In patients with intermediate (ie, neither low nor high) probability of lateralized aldosterone production, subsequent evaluation is guided by the individual's treatment preferences. For those who wish to consider surgical management, an aldosterone suppression test should be performed next to determine whether subtype classification is warranted. (See '[Aldosterone suppression testing](#)' below.)

In patients who are not surgical candidates or would not consider surgery, MRA therapy should be initiated. (See "[Treatment of primary aldosteronism](#)", section on '[Medical therapy](#)'.)

- **Consider medication effects** – Beta-adrenergic blockers and centrally acting alpha-2 agonists can decrease renin and potentially lead to false-positive results. However, testing results generally can be interpreted despite this potential interference. For example, although beta-adrenergic blockers lower PRA and DRC concentrations and raise the PAC/PRA ratio [25], these effects are **not** clinically important in this setting because patients without primary aldosteronism will not have an elevated PAC, whereas in patients with primary aldosteronism, an elevated PAC will still be evident [25].



In addition, the risks of modifying antihypertensive medication regimens (eg, hypertensive crisis, severe hypokalemia, atrial fibrillation, heart failure) often argue against even temporary discontinuation of therapy [30].

Negative findings on initial testing — If findings on initial testing are negative, potential causes of false-negative results should be considered before excluding the diagnosis. If clinical suspicion for primary aldosteronism is high, we suggest repeat testing given the high degree of intraindividual variability in these analytes in individuals both with and without primary aldosteronism [31,32]. False-negative results may occur

in the setting of hypokalemia or medications that raise renin (eg, MRAs, diuretics, angiotensin-converting enzyme [ACE] inhibitors, angiotensin receptor blockers [ARBs]).

- **Correct hypokalemia** – If hypokalemia is present, serum potassium should be corrected, and case-detection testing should be repeated on another day.
- **Assess for interfering drugs** – Potentially interfering medications should be considered in the interpretation of negative case-detection testing. Withdrawal of such medications for two to four weeks with retesting may be reasonable ([algorithm 1](#)). However, the risks of modifying antihypertensive medication programs (eg, hypertensive crisis, severe hypokalemia, atrial fibrillation, heart failure) often argue against even temporary discontinuation of therapy [30].

Potentially clinically important issues exist with the following drugs:

- **MRAs** – Data obtained from patients treated with an MRA (**spironolactone and eplerenone**) may be difficult to interpret. These drugs prevent aldosterone from activating its receptor, resulting sequentially in sodium loss, a decrease in plasma volume, and an elevation in PRA, which could potentially lead to false-negative testing in a patient with primary aldosteronism. For this reason, [spironolactone](#) and [eplerenone](#) should not be initiated until the evaluation is completed and the final decisions about treatment are made.

However, there are exceptions to this rule. For example, if the patient has hypokalemia despite treatment with [spironolactone](#) or [eplerenone](#), then the mineralocorticoid receptors are not fully blocked, and PRA or PRC should remain suppressed in patients with primary aldosteronism. Most patients with primary aldosteronism who are treated with MRAs are given subtherapeutic doses. Thus, when there is clinical suspicion for primary aldosteronism in patients treated with spironolactone or eplerenone, PAC and PRA should be measured; if the PRA is suppressed in this setting, these medications are not interfering with the evaluation, and case-detection testing, aldosterone suppression testing, and adrenal vein sampling (AVS) can be performed without discontinuing the MRAs [33,34]. However, if PRA is not suppressed, then the MRA should be discontinued for four to six weeks before retesting. Other potassium-sparing diuretics, such as [amiloride](#) and [triamterene](#), usually do not interfere with testing unless the patient is on high doses.

- **ACE inhibitors, ARBs, direct renin inhibitors** – ACE inhibitors, ARBs, and direct renin inhibitors could potentially elevate PRC and have variable effects on PRA in patients with primary aldosteronism. Thus, in a patient treated with one of these drugs, a PRA >1 ng/mL/hour does **not** exclude the diagnosis of primary aldosteronism. On the other hand, positive results on case-detection testing are diagnostic of primary aldosteronism. (See '[Test interpretation](#)' above.)

Other causes of hypertension and hypokalemia



Secondary hyperaldosteronism – Secondary hyperaldosteronism (eg, renovascular disease) should be considered when both the PRA (or PRC) and PAC are increased and the PAC/PRA ratio is <10. In this setting, hypersecretion of renin leads sequentially to increased angiotensin II and then increased aldosterone secretion.

An increased PRA or PRC in a patient with hypokalemia and hypertension is most often due to diuretic therapy, which may be surreptitious. Less common causes include renovascular or malignant hypertension and rare

renin-secreting tumors. Patients with renin-secreting tumors are typically young (average age 22 years in one report) and have severe hypertension and hypokalemia [35]. Contrast-enhanced computed tomography (CT) appears to be the diagnostic procedure of choice since false-negative results occur with arteriography or renal vein renin sampling. Surgical removal of the tumor cures the disease.

- **Liddle syndrome** – Liddle syndrome is a rare autosomal dominant condition in which there is a primary increase in sodium reabsorption in the collecting tubules and, in most cases, potassium secretion. The underlying defect is a gain-of-function variant in the collecting tubule sodium channel. Hypokalemia, hypertension, and low renin are common findings, but, in contrast with primary aldosteronism, plasma aldosterone concentrations are low. The evaluation and management of Liddle syndrome are reviewed separately. (See "[Genetic disorders of the collecting tubule sodium channel: Liddle syndrome and pseudohypoaldosteronism type 1](#)", section on 'Liddle syndrome'.)
- **Cushing syndrome** – Hypercortisolism of Cushing syndrome can overwhelm the ability of renal 11-beta-hydroxysteroid dehydrogenase type 2 to inactivate cortisol to cortisone. This allows cortisol to activate the mineralocorticoid receptor and cause hypertension and hypokalemia. Corticotropin (ACTH)-mediated causes of Cushing syndrome further lead to hypersecretion of ACTH-dependent mineralocorticoids such as deoxycorticosterone and corticosterone. With ectopic ACTH syndrome, ACTH and cortisol hypersecretion are much higher than with an ACTH-secreting pituitary adenoma, leading to a greater prevalence of hypokalemia (50 versus 9 percent in one series) [36]. (See "[Epidemiology and clinical manifestations of Cushing syndrome](#)" and "[Establishing the cause of Cushing syndrome](#)" and "[Apparent mineralocorticoid excess syndromes \(including chronic licorice ingestion\)](#)".)
- **Non-aldosterone mineralocorticoid excess** – The combination of suppressed PRA or PRC and a **low** plasma or urinary aldosterone value in a patient with hypertension and hypokalemia may indicate the presence of some non-aldosterone mineralocorticoid. In addition to Cushing syndrome, these findings can be evident in the following settings:
 - Some types of congenital adrenal hyperplasia (deficiencies of 11-beta-hydroxylase [CYP11B1, P450c11] or 17-alpha-hydroxylase [CYP17, P450c17]) cause hypertension and hypokalemia because of hypersecretion of the mineralocorticoid deoxycorticosterone. Familial cortisol resistance has a similar presentation. (See "[Adrenal steroid biosynthesis](#)".)
 - A deoxycorticosterone-producing adrenal tumor, which can usually be detected by CT or magnetic resonance imaging (MRI), also leads to non-aldosterone mineralocorticoid excess [1].
 - Other causes of mineralocorticoid excess include chronic licorice root ingestion and the rare genetic syndrome of apparent mineralocorticoid excess. (See "[Apparent mineralocorticoid excess syndromes \(including chronic licorice ingestion\)](#)".)



role for urine potassium measurement — We do **not** order a 24-hour urine potassium collection for the evaluation of primary hyperaldosteronism; the utility of this test is largely limited to clinical suspicion of surreptitious vomiting or laxative abuse. The low serum potassium concentration induced by mineralocorticoid excess is a result of increased urinary potassium excretion. Thus, in the past, a 24-hour urine collection was typically obtained to document the presence of inappropriate potassium wasting, defined as more than 30 mEq/day in a patient with hypokalemia. An appropriately low rate of potassium excretion may be evident in the

setting of either extrarenal losses (vomiting, diarrhea) or active diuretic treatment with urine collected during a nadir of the diuretic effect.

Interpretation of the rate of potassium excretion requires attention to the patient's volume status and rate of sodium excretion. The patient must not have a low sodium intake or hypovolemia (as evidenced by less than 50 mEq of sodium being excreted per day), since the associated decrease in sodium and water delivery to the distal potassium secretory site can diminish potassium excretion even in patients with hyperaldosteronism. On the other hand, the degree of potassium wasting and, therefore, the diagnostic accuracy, can be increased by a high-sodium diet because the combination of increased distal flow and hypersecretion of aldosterone will maximize potassium losses [37].

ALDOSTERONE SUPPRESSION TESTING

Indications — Aldosterone suppression testing is used to verify the diagnosis of primary aldosteronism and assess whether subtype classification (which frequently includes adrenal vein sampling [AVS]) should be pursued. It is indicated in individuals who have an intermediate probability of lateralized aldosterone production **and** are willing and able to undergo surgical management ([algorithm 2](#)) [6]. (See '[Positive findings on initial testing](#)' above.)

We agree with the 2025 Endocrine Society clinical practice guideline that aldosterone suppression testing is generally **not** needed in the following three settings:

- When a patient with positive case-detection testing does **not** want to consider surgical management.
- When the probability of lateralizing primary aldosteronism is high. (See '[Positive findings on initial testing](#)' above.)
- When the probability of lateralizing primary aldosteronism is so low that pursuing aldosterone suppression testing is not needed clinically. (See '[Positive findings on initial testing](#)' above.)

Aldosterone suppression testing can be performed with orally administered sodium chloride and measurement of urine aldosterone excretion or with intravenous sodium chloride loading and measurement of plasma aldosterone concentration (PAC) [22,38].

Oral sodium loading — Many centers and experts, including the author of this topic, use oral sodium loading over three days [39]. After hypertension and hypokalemia are controlled (hypokalemia suppresses aldosterone secretion), the patient should receive a high-sodium diet for three days.

Testing protocol – Patients should be given guidance on the sodium content of the types of food they need to consume to achieve a 5000 mg sodium diet. Alternatively, particularly if patients have a taste aversion to high-sodium foods, they can be given [oral sodium chloride](#) tablets (eg, two 1 g sodium chloride tablets taken three times daily with food will provide approximately 90 mEq of sodium). On the third day of the high-sodium diet, serum electrolytes are measured, and a 24-hour urine specimen is collected for measurement of aldosterone, sodium, and creatinine. The 24-hour urine sodium excretion should exceed 200 mEq (4600 mg) to document adequate sodium loading. Urine aldosterone excretion >12 mcg/24 hours (33 nmol/day) in this setting is consistent with hyperaldosteronism. (See "[Patient education: Collection of a 24-hour urine specimen \(Beyond the Basics\)](#)".)

During oral sodium loading, the serum potassium level should be measured daily. Sodium-induced hypokalemia itself is strongly suggestive of nonsuppressible hyperaldosteronism. Healthy individuals do not waste potassium during sodium loading, because the increase in distal flow is offset by reduced secretion of aldosterone.

- **Risks and safety precautions** – The risk of increasing dietary sodium in patients with severe hypertension must be assessed for each individual. In addition, since sodium loading typically increases kaliuresis and hypokalemia, patients are at risk for severe hypokalemia. Vigorous replacement of [potassium chloride](#) should be prescribed as indicated.

Saline infusion test — An alternate method to suppress endogenous aldosterone production is by the intravenous administration of 2 L of isotonic [saline](#) over four hours (from 8 AM to 12 PM), ideally while the patient is seated [38,40]. The PAC will fall below 5 ng/dL (139 pmol/L) in healthy individuals, whereas values ≥ 10 ng/dL (277 pmol/L) are consistent with primary aldosteronism [38]. False-negative rates may be as high as 30 percent, but they appear to be lower if the test is performed with the patient seated rather than recumbent [40,41].

Other — Other available confirmation tests include the [fludrocortisone](#) suppression and [captopril](#) challenge tests. As noted above, we prefer oral sodium loading. (See 'Oral sodium loading' above.)

SUBTYPE CLASSIFICATION

Purpose — Subtype classification is only needed in patients with an established diagnosis of primary aldosteronism who want to pursue surgical management. The purpose is to distinguish unilateral aldosterone-producing adenoma (APA), or rarely, carcinoma, from bilateral adrenal hyperplasia (idiopathic hyperaldosteronism [IHA]). This distinction is important since the treatment options are different for the two disorders. To differentiate APA from IHA, we employ an algorithm that uses adrenal CT and adrenal vein sampling (AVS), consistent with the 2025 Endocrine Society clinical practice guideline ([algorithm 2](#)) [6]. (See "Treatment of primary aldosteronism".)

- **APA** – In general, patients with APA have higher aldosterone secretion rates, resulting in more severe hypertension, more profound hypokalemia (< 3.2 mEq/L), and higher plasma (> 20 ng/dL) and urinary (> 20 mcg/24 hour) levels of aldosterone; these patients are also typically younger (age < 50 years) than those with IHA [22]. In one study, a plasma aldosterone concentration (PAC)/plasma renin activity (PRA) ratio of > 32 had a sensitivity of 100 percent and specificity of 61 percent for an APA [23].

The genetic basis of APA can be established in a substantial number of patients. Somatic genetic variants in *KCNJ5*, *ATP1A1*, *ATP2B3*, *CTNNB1*, and *CACNA1D* are found in more than 80 percent of resected APAs [42-45]. In a study of 474 unselected patients with APAs, somatic heterozygous *KCNJ5* variants were present in 38 percent of patients; *CACNA1D*, *ATP1A1*, or *ATP2B3* variants were identified in 16.3 percent [43]. Nonetheless, the presence of one of these somatic variants does not affect diagnosis or treatment. (See "Pathophysiology and clinical features of primary aldosteronism", section on 'Genetic variants in aldosterone-producing adenomas' and 'Familial hyperaldosteronism' below.)

- **IHA** – Bilateral adrenal hyperplasia, which accounts for approximately 60 percent of cases, is generally a milder disease with less hypersecretion of aldosterone and less frequent hypokalemia; it should be treated with a

mineralocorticoid receptor antagonist (MRA). The genetic basis of primary aldosteronism due to bilateral adrenal hyperplasia has not yet been determined. (See "[Treatment of primary aldosteronism](#)".)

- **Co-existing APA and IHA** – The clear distinction between unilateral aldosteronoma and bilateral adrenal hyperplasia has been challenged by the demonstration of the frequent presence of zona glomerulosa hyperplasia and aldosterone-producing micronodules adjacent to the dominant aldosteronoma. In addition, while ion channel genetic variants are frequently found in the dominant aldosteronoma, they are not present in the adjacent hyperplasia, suggesting that somatic development of a dominant adenoma may occur in a background of bilateral mild hyperplasia [46,47].

Adrenal CT — Adrenal computed tomography (CT) should be the initial study to determine subtype (adenoma versus hyperplasia) and exclude adrenal carcinoma [48]. For adrenal gland imaging, CT has superior spatial resolution compared with MRI. The CT scan may be done without intravenous contrast; however, if an adrenal mass is detected, contrast administration provides additional imaging information. (See "[Evaluation and management of the adrenal incidentaloma](#)", section on 'Typical imaging features'.)

Interpretation

- An adrenal carcinoma should be suspected when a unilateral, large (>4 cm) adrenal mass is found on CT in a patient with primary aldosteronism ([image 1](#)) [49-53].
- Bilateral adrenal gland thickening or micronodular changes suggest adrenal hyperplasia; however, patients with hyperplasia may also have normal-appearing adrenal glands on CT [50].
- When a solitary, hypodense, unilateral macroadenoma (>1 cm and <2 cm) and normal contralateral adrenal morphology ([image 2](#)) are found in a young patient (<35 years of age) with severe primary aldosteronism, unilateral adrenalectomy without prior AVS is generally reasonable and informed by shared decision-making with the patient [6,54]. However, because of the age-dependent risk that a solitary unilateral adrenal macroadenoma may be a nonfunctioning cortical adenoma, AVS is preferred if resources and expertise are available for patients aged ≥35 years who want to pursue a surgical cure of hyperaldosteronism ([algorithm 2](#)). (See 'Indications' below.)
- For macroadenomas ≥1.5 cm that likely represent an APA, assessment for cortisol cosecretion is warranted, as this informs subsequent management decisions. (See 'Cortisol cosecretion' below.)

Limitations — Some investigators suggest that the findings of hypokalemia, nonsuppressible hyperaldosteronism, a PAC/PRA ratio >50, and a unilateral mass on CT can be followed directly by surgery to remove a presumed adenoma [28]. However, CT findings are frequently misleading, as many patients with biochemical evidence of nonsuppressible hyperaldosteronism and a unilateral adrenal mass turn out to have bilateral adrenal hyperplasia [51,53,55-57].

Another problem is that the absence of a mass does not exclude an adenoma since APAs can be very small (eg, <3 mm in diameter), and lesions <1 cm in diameter may be missed on CT. Further, bilateral lesions are not diagnostic of hyperplasia because some patients with an aldosteronoma in one adrenal gland have a nonfunctioning adrenal nodule in the other [52,53].

The limitations of adrenal CT were illustrated in a study of 203 patients with primary aldosteronism who were evaluated with both CT and AVS; CT was accurate in only 53 percent of patients [56]. Based upon CT, 42 patients

(22 percent) would have been incorrectly excluded as candidates for adrenalectomy, and 48 (25 percent) might have had unnecessary or inappropriate surgery. In another study, CT findings coincided with the lateralization determined by AVS in only 80 of 158 (51 percent) patients [58].

In a subsequent systematic review of 38 studies in a total of 950 patients with primary aldosteronism (including the 203 patients described above), adrenal CT/MRI results did not agree with AVS in 359 of 950 patients (37.8 percent) [57]. If CT/MRI alone had been used to determine subtype, the following inappropriate treatment would have been recommended:

- 139 patients (14.6 percent) would have inappropriately undergone unilateral adrenalectomy (unilateral mass on CT/MRI but bilateral findings on AVS), which would not have been curative.
- 181 patients (19.1 percent) would have been offered medical therapy instead of curative adrenalectomy (bilateral findings on CT/MRI but unilateral secretion on AVS).
- 37 patients (3.9 percent) would have undergone adrenalectomy on the wrong side (AVS showing unilateral secretion on the opposite side of CT/MRI abnormalities).

These observations highlight the importance of performing AVS in most patients to distinguish between unilateral and bilateral adrenal aldosterone hypersecretion. (See '[Adrenal vein sampling](#)' below.)

The role of CT and MRI in the evaluation of incidental adrenal masses is reviewed in detail elsewhere. (See '[Evaluation and management of the adrenal incidentaloma](#)', section on 'MRI'.)

Adrenal vein sampling — Measurement of aldosterone in samples of adrenal venous blood, obtained by an experienced interventional radiologist, is the criterion standard test (ie, gold standard) to distinguish between unilateral adenoma (or unilateral hyperplasia) and bilateral adrenal hyperplasia [59]. Lateralized disease is associated with a marked (usually fourfold greater than contralateral adrenal) increase in PAC on the side of the tumor, whereas there is little difference between the two sides in patients with bilateral hyperplasia ([figure 1](#) and [figure 2](#)) [56,60,61].

Performance of AVS by experienced personnel is critical to optimize test success and minimize complications. One limitation of AVS is an inability to obtain good samples as the right adrenal vein is small and sometimes difficult to locate; due to this and other technical challenges, success rates vary in part with local expertise. In two series, catheterization was successful in 43 of 49 patients (88 percent) and 194 of 203 patients (96 percent), respectively ([figure 2](#)) [55,56]. The complication rate in published studies of AVS is 2.5 percent or less [56,62-65]. The most common complication is groin hematoma; adrenal hemorrhage and adrenal vein dissection are rare [66].

Indications — For patients aged ≥ 35 years who would like to pursue surgical management (unilateral adrenalectomy) of primary aldosteronism, we suggest AVS to confirm unilateral disease regardless of CT findings [54]. If AVS is not available and the patient cannot travel to a medical center with AVS expertise, medical management with an MRA is a reasonable option. Alternatively, for patients with severe primary aldosteronism (ie, high probability of lateralizing disease) **and** a unilateral adrenal nodule, unilateral adrenalectomy may be pursued. However, if the surgical option is selected, it is important for both patient and provider to understand that the adrenal nodule may be a nonfunctioning adenoma and that surgery may therefore not be curative.

We suggest that AVS may not be needed in patients aged <35 years who have severe primary aldosteronism (ie, high probability of lateralizing disease) and a unilateral adrenal macroadenoma (>1 cm and <2 cm), because they are unlikely to have a nonfunctioning adrenal adenoma that could be confused with an APA ([algorithm 2](#)) [67]. However, APAs are generally small tumors, so when an adenoma >2 cm in size is found in a patient with primary aldosteronism, it is typically not aldosterone secreting. The development of adrenocortical nodularity is, in part, a function of age; therefore, the likelihood that an adenoma is an APA rather than a nonfunctioning adrenal tumor is higher in younger (ie, aged <35 years) than in older patients with primary aldosteronism.

The biochemical criteria for high probability of APA are reviewed above. (See '[Positive findings on initial testing](#)' above.)

Procedure — Some centers perform AVS without cosyntropin stimulation, but we prefer **continuous cosyntropin infusion** (50 mcg per hour started 30 minutes before sequential sampling of the adrenal veins and continued throughout the procedure) for the following reasons:

- To minimize stress-induced fluctuations in aldosterone secretion during nonsimultaneous AVS, which could potentially confound the interpretation of lateralization data.
- To maximize the gradient in cortisol from adrenal vein to inferior vena cava (IVC) and thus confirm successful sampling of the adrenal vein.
- To maximize the secretion of aldosterone from an APA [56,60].

Some investigators have suggested that when given as a bolus injection and when the adrenal veins are sampled simultaneously, cosyntropin administration does not improve the diagnostic accuracy of AVS and may misclassify some patients [61,62,68,69]. However, as noted above, we suggest **continuous cosyntropin infusion** for optimal results [56,60,70,71]. Confidence in successful cannulation of both adrenal veins is critical to patient care. If the clinician cannot be confident that both adrenal veins were successfully sampled, the AVS data are not clinically useful.

Aldosterone and cortisol concentrations are measured in the blood from all three sites (right adrenal vein, left adrenal vein, and IVC). All of the blood samples should be assayed at 1:1, 1:10, and 1:50 dilutions; absolute values and accurate laboratory assays for cortisol and aldosterone are essential for successful interpretation of the AVS data. An AVS-specific report should be developed by the laboratory to prevent any confusion on data interpretation ([figure 2](#)) [22,56,63].

Confirming successful catheterization — The cortisol concentrations from the adrenal veins and IVC are used to confirm successful cannulation of both adrenal veins. With cosyntropin infusion, the adrenal vein-to-IVC cortisol ratio is typically more than 10:1 [56]; a ratio of at least 5:1 is required to be confident that the adrenal veins were successfully catheterized ([figure 2](#)).

However, when cosyntropin infusion is not used, an adrenal vein-to-IVC cortisol gradient of more than 3:1 is recommended as the threshold to verify successful catheterization [64]. Some centers require only a 10 percent gradient between an adrenal vein and the IVC [61,62], a change that can be seen in minute-to-minute adrenal cortisol secretion and that is within the coefficient of variation of some cortisol assays. Thus, as noted, we suggest cosyntropin infusion during AVS.

Cortisol-corrected ratios — Dividing the right and left adrenal vein PAC by their respective cortisol concentrations corrects for the dilutional effect of the inferior phrenic vein flow into the left adrenal vein; these are termed "cortisol-corrected ratios."

- At Mayo Clinic, where AVS is performed with cosyntropin infusion, the mean cortisol-corrected aldosterone ratio (APA-side PAC/cortisol to normal adrenal PAC/cortisol) in patients with confirmed APA is 18:1 [56]. We use a cutoff for the cortisol-corrected aldosterone ratio from high-side to low-side of more than 4:1 to indicate unilateral aldosterone excess [56].
- In patients with presumed IHA, the mean cortisol-corrected aldosterone ratio is 1.8:1 (high-side to low-side); a ratio less than 3:1 is suggestive of bilateral aldosterone hypersecretion [56].

Thus, most patients with a unilateral source of aldosterone will have cortisol-corrected aldosterone lateralization ratios greater than 4; ratios greater than 3, but less than 4, represent a zone of overlap. A ratio less than 3 is consistent with bilateral aldosterone hypersecretion [56].

In addition, the contralateral aldosterone-to-cortisol ratio is less than the IVC aldosterone-to-cortisol ratio in 93 percent of patients with surgically confirmed APA [56], indicating suppression of aldosterone secretion by the noninvolved adrenal gland.

Centers that perform AVS without cosyntropin infusion use lower lateralization cutoff values [61,62]. However, using the diagnostic cutoffs described above for cosyntropin-stimulated AVS, the sensitivity and specificity for detecting unilateral aldosterone hypersecretion are 95 and 98.6 percent, respectively [54]. These test characteristics deteriorate when lower cutoffs are used to determine successful catheterization and lateralization.

Special circumstances — AVS may be most useful when no adrenal abnormality is found on CT or when both adrenal glands are abnormal but asymmetric. In one report, for example, 24 of 58 patients (41 percent) with normal adrenal CT and 16 of 33 (49 percent) with bilateral micronodules on CT had a unilateral source of aldosterone by AVS [56].

There are isolated cases of primary aldosteronism due to an ectopic adrenal adenoma (as in the kidney) [72]. These patients have low serum aldosterone concentrations in AVS, and CT or MRI may identify the site of the tumor.

Tests not recommended

- **Limited evidence** – A long-term goal should be to decrease the reliance on specialized interventional radiologists for AVS. Initial studies suggest that molecular imaging with positron emission tomography (PET) may provide a noninvasive alternative to AVS [73]. For example, in a study of 174 participants with primary aldosteronism seeking subtype classification for possible adrenalectomy, participants underwent both AVS and post-dexamethasone [¹¹C]metomidate ([¹¹C]MTO) PET/CT [74]. AVS and [¹¹C]MTO-PET/CT performed similarly in predicting complete or partial biochemical response after adrenalectomy (62.8 percent, 95% CI 52.2-72.6 percent and 71.3 percent, 95% CI 61.0-80.1 percent, respectively).
- **Limited utility** – Other tests that predate the preferred approach with adrenal CT and AVS were used to distinguish unilateral APAs from bilateral disease but have limited clinical utility. These tests include:

- **Posture stimulation test** – The posture stimulation test was based upon the observation that patients with bilateral adrenal hyperplasia have a characteristic rise in PAC when going from the supine to standing position; this phenomenon has been ascribed to an enhanced sensitivity of the adrenal zona glomerulosa to the small changes in angiotensin II that occur with standing [75]. In contrast, no such changes would be expected in patients with an APA because their aldosterone hypersecretion is autonomous and diurnal.

However, this test does not discriminate well between unilateral adenoma and bilateral hyperplasia [7,49,76,77]. As an example, in a study of 20 patients with primary aldosteronism (15 with a unilateral APA) undergoing a postural stimulation test, PAC increased after four hours of standing in all patients with hyperplasia and in 8 of 15 patients with adenomas (based upon a 30 percent rise) [49].

- **18-hydroxycorticosterone** – Patients with an APA typically have elevated supine plasma 18-hydroxycorticosterone levels at 8 AM (>100 ng/dL), whereas patients with bilateral IHA do not [48]. However, the accuracy of the test is low, and it does not help with localization [48,77].
- **Iodocholesterol scintigraphy** – Radionuclide scintigraphy with ¹³¹I-iodocholesterol or an analog (NP-59) is no longer used in most centers. While it has the potential advantage of correlating function with anatomic findings, it is not useful for evaluating small adrenal nodules, as tracer uptake is poor in APAs <1.5 cm in diameter [78]. This imaging modality is no longer available in the United States.

Familial hyperaldosteronism — Four rare forms of familial hyperaldosteronism (FH) are associated with adrenal hyperplasia:

- FH type I or glucocorticoid-remediable aldosteronism (GRA) due to a *CYP11B1/CYP11B2* chimeric gene
- FH type II, caused by germline pathogenic variants in the chloride channel *CLCN2* [79]
- FH type III, caused by germline pathogenic variants in the potassium channel subunit *KCNJ5*
- FH type IV, caused by germline pathogenic variants in the *CACNA1H* gene, which encodes the alpha subunit of an L-type voltage-gated calcium channel (Cav3.2)

Features suggesting FH include age <20 years and a family history of early-onset hypertension or primary aldosteronism. The decision to perform AVS depends on the specific form of FH and is discussed separately. (See "Familial hyperaldosteronism".)

Primary bilateral macronodular adrenal hyperplasia — Primary bilateral macronodular adrenal hyperplasia (PBMAH) is most frequently associated with excess glucocorticoid secretion. However, in those patients with cosecretion of cortisol and aldosterone, the disease is bilateral, and AVS is not needed. PBMAH is reviewed in detail separately. (See "Cushing syndrome due to primary bilateral macronodular adrenal hyperplasia".)



CORTISOL COSECRETION

Importantly, in a patient with a large aldosterone-producing adenoma (APA) cosecretion of cortisol may occur in patients with larger APAs (eg, ≥1.5 cm in diameter) [80]. It is reasonable to screen for this possibility in patients with apparent APAs ≥1.5 cm in diameter by measuring a baseline serum dehydroepiandrosterone sulfate (DHEAS) and performing a 1 mg overnight dexamethasone suppression test.

If autonomous cortisol cosecretion is present in a patient with primary aldosteronism and a unilateral adrenal macroadenoma, then adrenal vein sampling (AVS) is not needed. In this setting, the macroadenoma is usually

cosecreting aldosterone and cortisol. However, even if the patient has primary aldosteronism caused by bilateral idiopathic hyperaldosteronism (IHA) or a contralateral adrenal micronodule, adrenalectomy remains warranted for management of hypercortisolism because good, long-term medical treatment options are not available for hypercortisolism. In contrast, if primary aldosteronism persists following unilateral adrenalectomy, mineralocorticoid receptor antagonists (MRAs) are an excellent treatment. When autonomous cortisol cosecretion is documented, patients must be treated perioperatively with stress doses of glucocorticoids followed by a planned postoperative taper. (See ["Overview of the treatment of Cushing syndrome", section on 'Adrenal adenomas'](#).)

PREGNANCY

Primary aldosteronism is uncommon in pregnancy, with fewer than 60 patients reported in the medical literature; most of these patients have had aldosterone-producing adenomas (APAs) [81-85]. Primary aldosteronism can lead to intrauterine growth retardation, preterm delivery, intrauterine fetal demise, and placental abruption [85,86].

- **Case detection** – Case-detection testing for primary aldosteronism during pregnancy is the same as for nonpregnant patients: morning blood sample for the measurement of plasma aldosterone concentration (PAC) and plasma renin activity (PRA) or renin concentration (PRC). Suppressed renin and an aldosterone level ≥ 10 ng/dL (277 pmol/L) is a positive case-detection test for primary aldosteronism. If spontaneous hypokalemia is present with high aldosterone (>20 ng/dL [554 pmol/L]) and suppressed renin, aldosterone suppression testing is not needed. In patients with normokalemia and a positive case-detection test, aldosterone suppression testing should be pursued. However, the [captopril](#) stimulation test is contraindicated in pregnancy, and the [saline](#) infusion test may not be well tolerated. One option is measurement of sodium and aldosterone in a 24-hour urine collection on an ambient sodium diet. (See ["Initial testing"](#) above and ["Aldosterone suppression testing"](#) above.)
- **Subtype classification** – Subtype classification with abdominal MRI without gadolinium is the test of choice. Adrenal imaging with CT, iodocholesterol scintigraphy, and adrenal vein sampling (AVS) should be **avoided** in pregnancy. As highlighted in the 2025 Endocrine Society clinical practice guideline on primary aldosteronism [6], AVS may not be needed in patients with severe primary aldosteronism (ie, spontaneous hypokalemia and PAC >20 ng/dL [554 pmol/L]) and age <35 years who have a clear-cut, unilateral adrenal adenoma on cross-sectional imaging [54]. (See ["Subtype classification"](#) above and ["Treatment of primary aldosteronism", section on 'Pregnancy'](#).)

ENDOCRINE SOCIETY GUIDELINE LINKS



Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Primary aldosteronism"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topic (see ["Patient education: Aldosterone test \(The Basics\)"](#))

SUMMARY AND RECOMMENDATIONS

- **Causes of primary aldosteronism** – The most common causes of primary aldosteronism are aldosterone-producing adenomas (APAs) and bilateral idiopathic hyperaldosteronism (IHA). Other, rarer causes include familial hyperaldosteronism (FH) type I (glucocorticoid-remediable aldosteronism [GRA]), type II, type III, or type IV. (See ["Introduction"](#) above and ["Familial hyperaldosteronism"](#).)
- **Prevalence and presentation** – The prevalence of primary aldosteronism is considerably higher than previously thought. The classic presenting signs of primary aldosteronism are hypertension and hypokalemia. However, normokalemia is more common than hypokalemia in patients diagnosed with primary aldosteronism. (See ["Prevalence"](#) above and ["Variable presentation"](#) above.)
- **Case detection** – Consistent with the 2025 Endocrine Society guidelines, the author of this topic performs testing for primary aldosteronism in all patients with hypertension ([algorithm 1](#)). Identifying primary aldosteronism is important because of both its high prevalence and association with higher cardiovascular morbidity and mortality compared with primary hypertension at the same degree of blood pressure elevation. (See ["Patient selection"](#) above.)
- **Initial testing** – For initial testing, paired measurements of the plasma renin activity (PRA; or direct renin concentration [DRC]), plasma aldosterone concentration (PAC), and serum potassium should be obtained, preferably in the morning (eg, 8:00 AM) in a seated, ambulatory patient. (See ["Initial testing"](#) above.)

In patients with primary aldosteronism (see ["Test interpretation"](#) above):

- The PRA is usually ≤ 1 ng/mL/hour (0.2778 ng/L per sec), and the DRC is below the lower limit of normal (typically ≤ 8.2 mU/L).
- The PAC is usually ≥ 10 ng/dL (277 pmol/L) when measured by immunoassay or ≥ 7.5 ng/dL (208 pmol/L) when measured by liquid chromatography-tandem mass spectrometry (LC-MS/MS).
- The PAC/PRA ratio is usually >20 ng/dL per ng/mL/hour (>555 pmol/L per ng/mL/hour) when aldosterone is measured by immunoassay or >15 ng/dL per ng/mL/hour (>416 pmol/L per ng/mL/hour) when measured

by LC-MS/MS ([table 1](#)). However, we prefer to use the paired random PAC and PRA (or DRC) for case detection rather than the PAC/PRA ratio. (See '[Aldosterone-to-renin ratio](#)' above.)

- **Subsequent evaluation** – In patients with findings consistent with primary aldosteronism on initial testing, the likelihood of lateralized aldosterone production (ie, APA) should be assessed ([algorithm 2](#)). If findings on initial testing are negative, potential causes of false-negative results should be considered before excluding the diagnosis. (See '[Subsequent evaluation](#)' above.)
- **Aldosterone suppression testing** – Aldosterone suppression testing is indicated for patients who have an intermediate probability of lateralized aldosterone production **and** are willing and able to undergo surgical management. It is not needed in patients with either a high or low probability of lateralizing disease or in those who would not consider adrenalectomy. (See '[Aldosterone suppression testing](#)' above.)

For aldosterone suppression testing, we use oral sodium loading and measurement of urine aldosterone excretion. Some experts prefer intravenous sodium chloride loading and measurement of the PAC.

- **Subtype classification** – Subtype classification is only needed in patients with an established diagnosis of primary aldosteronism who would consider surgical management ([algorithm 2](#)). The purpose is to distinguish unilateral APA, or rarely, carcinoma, from IHA).
- **Adrenal CT** – We suggest adrenal CT as the initial test to distinguish between APA and bilateral hyperplasia. Adrenal CT will also exclude adrenocortical carcinoma. (See '[Adrenal CT](#)' above.)

For macroadenomas ≥ 1.5 cm that likely represent an APA, assessment for cortisol cosecretion is warranted, as this informs subsequent management decisions. (See '[Cortisol cosecretion](#)' above.)

- **Adrenal vein sampling (AVS)** – For patients aged ≥ 35 years who wish to pursue surgical management, we recommend AVS to confirm unilateral disease regardless of CT findings ([figure 2](#)).

AVS may not be needed in patients aged < 35 years who have severe primary aldosteronism and a unilateral adrenal macroadenoma (> 1 cm and < 2 cm). (See '[Adrenal vein sampling](#)' above.)

AVS should only be performed by an experienced radiologist. (See '[Adrenal vein sampling](#)' above.)

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